
Productivity and Misallocation

Main paper: Hsieh and Klenow (2009). Related papers: Olley–Pakes (1996), Restuccia–Rogerson (2008), Asker–Collard-Wexler–De Loecker (2014).

Opening puzzle

Can productivity rise without new technology, more capital, or more workers?

Yes: if resources move to the right producers.

Productive but constrained plant

A_i high, $MRPK_i$ high

- high physical productivity,
- high marginal return to inputs,
- too little capital and labor.

Less productive but favored plant

A_j low, $MRPK_j$ low

- subsidized credit or protection,
- low marginal return,
- too much capital and labor.

Misallocation: resources are not where their marginal products are highest.

Roadmap: HK is the center

Main question: How large are TFP losses when productive plants are too small?

Paper	Connection to HK	Role in this presentation
Olley–Pakes (1996)	Productivity growth can come from reallocation.	Empirical motivation: reallocation matters.
Restuccia–Rogerson (2008)	Idiosyncratic distortions can lower aggregate TFP.	Theory foundation: distortions cause misallocation.
Hsieh–Klenow (2009)	TFPR dispersion measures marginal-product gaps.	Main framework: misallocation accounting.
Asker et al. (2014)	Dynamic capital adjustment can create MRPK dispersion.	Caveat: dispersion is a signal, not automatic proof.

OP motivates HK; RR explains HK's mechanism; Asker refines HK's interpretation.

The one-chain view

plant heterogeneity $\Rightarrow$ marginal-product gaps $\Rightarrow$ TFPR dispersion $\Rightarrow$ aggregate TFP loss.

Static efficiency condition

$$\text{MRPK}_{si} = \text{MRPK}_{sj}, \quad \text{MRPL}_{si} = \text{MRPL}_{sj}.$$

- If marginal products differ, moving inputs can raise output.
- HK makes this condition measurable with TFPR.

Dynamic interpretation caveat

$$K_{t+1} = (1 - \delta)K_t + I_t.$$

- Capital is slow to adjust.
- Some observed MRPK gaps may be dynamic, not policy-driven.

Message: observed dispersion is a diagnostic signal; the model determines the interpretation.

Part I. Why HK is motivated

Reallocation matters empirically, and distortions can block it

Olley–Pakes: reallocation matters empirically

Question. Did productivity growth after deregulation come from average plant improvement, or from reallocation toward more productive plants?

OP decomposition

$$P_t = \sum_i s_{it} p_{it} = \bar{p}_t + \sum_i (s_{it} - \bar{s}_t)(p_{it} - \bar{p}_t).$$

- $\bar{p}_t$: unweighted mean productivity.
- covariance term: whether productive plants have larger shares.

Interpretation

- If high- p plants gain output share, aggregate productivity rises.
- Productivity growth can come from reallocation, not only technology.

Restuccia–Rogerson: distortions can lower TFP

Question. Can producer-specific policies lower aggregate TFP even when aggregate technology is unchanged?

Heterogeneous establishment technology

$$f(s, k, n) = sk^\alpha n^\gamma, \quad 0 < \alpha + \gamma < 1.$$

Reading this equation. Establishments differ in productivity s . Capital k and labor n produce output with decreasing returns because $\alpha + \gamma < 1$.

Profit with an output distortion

$$\pi(s, \tau) = \max_{k, n} \{ (1 - \tau)sk^\alpha n^\gamma - rk - wn - c_f \}.$$

Interpretation. A positive τ acts like a tax; a negative τ acts like a subsidy. The distortion changes the producer's optimal scale.

Mechanism

- Taxing high-productivity establishments makes them too small.
- Subsidizing low-productivity establishments makes them too large.
- The correlation between productivity s and distortion τ is crucial.

Connection to HK

RR gives the theory experiment:

$(s, \tau) \Rightarrow$ distorted scale $\Rightarrow$ lower aggregate TFP.

HK measures the implied wedges from plant data:

plant data $\Rightarrow$ TFPR dispersion $\Rightarrow$ TFP loss.

Notation

$f(s, k, n)$	establishment output	s	establishment productivity	k	capital input
n	labor input	α	capital output elasticity	γ	labor output elasticity
$\alpha + \gamma < 1$	decreasing returns to scale	$\pi(s, \tau)$	establishment profit	$\max_{k, n}$	choose capital and labor optimally
τ	output tax or subsidy wedge	$1 - \tau$	effective output-revenue factor	r	rental price of capital
w	wage per unit of labor	c_f	fixed operating cost	TFP	total factor productivity

Part II. Hsieh and Klenow (2009)

The main measurement framework

HK research question and identification logic

Research question. How much would manufacturing TFP rise if resources were reallocated toward high-return plants?

What HK observes

- plant revenue,
- wage bill and capital stock,
- industry classification,
- China, India, and U.S. manufacturing data.

What HK infers

$$\text{data} \Rightarrow \text{MRPK}, \text{MRPL} \Rightarrow \text{TFPR} \Rightarrow \text{wedges} \Rightarrow \frac{Y}{Y^{\text{eff}}}.$$

This is model-based measurement, not a randomized causal design.

Key benchmark: compare China and India to U.S. dispersion, not to zero dispersion.

HK production structure

$$\underbrace{Y = \prod_{s=1}^S Y_s^{\theta_s}}_{\text{final good}}, \quad \underbrace{Y_s = \left(\sum_{i=1}^{M_s} Y_{si}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}}_{\text{industry CES}}, \quad \underbrace{Y_{si} = A_{si} K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}}_{\text{plant}}.$$

θ_s
industry expenditure share

σ
substitution elasticity across
plant varieties

α_s
capital share in industry s

Demand and wedges

CES demand

$$Y_{si} = \left(\frac{P_{si}}{P_s} \right)^{-\sigma} Y_s, \quad P_{si} = P_s \left(\frac{Y_s}{Y_{si}} \right)^{1/\sigma}.$$

As a plant expands, its price falls. This is why high-TFPQ plants do not necessarily have high TFPR in an efficient allocation.

Profit with distortions

$$\pi_{si} = (1 - \tau_{Ysi})P_{si}Y_{si} - wL_{si} - (1 + \tau_{Ksi})RK_{si}.$$

τ_Y affects both inputs through revenue; τ_K affects capital relative to labor.

Wedges are reduced-form objects: policy, credit, market access, markups, or measurement can all appear as wedges.

Deriving the first-order conditions

Suppress *si* subscripts inside the derivation.

$$\pi = (1 - \tau_Y)P(Y)Y - wL - (1 + \tau_K)RK, \quad Y = AK^\alpha L^{1-\alpha}.$$

$$P(Y)Y = P_s Y_s^{1/\sigma} Y^{1-1/\sigma} \Rightarrow MR \equiv \frac{\partial(PY)}{\partial Y} = \frac{\sigma-1}{\sigma} P.$$

Labor FOC

$$0 = \pi_L = (1 - \tau_Y)MR(1 - \alpha)\frac{Y}{L} - w.$$

$$\boxed{\text{MRPL}_{si} = (1 - \alpha_s) \frac{\sigma-1}{\sigma} \frac{P_{si} Y_{si}}{L_{si}} = \frac{w}{1 - \tau_{Ysi}}}$$

Capital FOC

$$0 = \pi_K = (1 - \tau_Y)MR\alpha\frac{Y}{K} - (1 + \tau_K)R.$$

$$\boxed{\text{MRPK}_{si} = \alpha_s \frac{\sigma-1}{\sigma} \frac{P_{si} Y_{si}}{K_{si}} = R \frac{1 + \tau_{Ksi}}{1 - \tau_{Ysi}}}$$

From FOCs to TFPR

Object. TFPR is plant revenue per capital–labor input bundle.

$$\text{TFPR}_{si} \equiv P_{si} A_{si} = \frac{P_{si} Y_{si}}{K_{si}^{\alpha_s} (wL_{si})^{1-\alpha_s}}, \quad \lambda \equiv \frac{\sigma - 1}{\sigma}.$$

1. Rewrite FOCs as revenue ratios

$$\frac{\text{MRPK}_{si}}{\alpha_s} = \lambda \frac{P_{si} Y_{si}}{K_{si}}, \quad \frac{\text{MRPL}_{si}}{w(1 - \alpha_s)} = \lambda \frac{P_{si} Y_{si}}{wL_{si}}.$$

2. Combine using input weights

$$\left(\frac{\text{MRPK}_{si}}{\alpha_s} \right)^{\alpha_s} \left(\frac{\text{MRPL}_{si}}{w(1 - \alpha_s)} \right)^{1-\alpha_s} = \lambda \text{TFPR}_{si}.$$

$$\text{TFPR}_{si} = \frac{1}{\lambda} \left(\frac{\text{MRPK}_{si}}{\alpha_s} \right)^{\alpha_s} \left(\frac{\text{MRPL}_{si}}{w(1 - \alpha_s)} \right)^{1-\alpha_s}$$

TFPR is a weighted geometric average of marginal revenue products.

Substituting wedges into TFPR

Starting point

$$\text{TFPR}_{si} = \frac{\sigma}{\sigma - 1} \left(\frac{\text{MRPK}_{si}}{\alpha_s} \right)^{\alpha_s} \left(\frac{\text{MRPL}_{si}}{w(1 - \alpha_s)} \right)^{1 - \alpha_s}.$$

1. Substitute the wedge FOCs

$$\text{MRPK}_{si} = R \frac{1 + \tau_{Ksi}}{1 - \tau_{Ysi}}, \quad \text{MRPL}_{si} = \frac{w}{1 - \tau_{Ysi}}.$$

$$\text{TFPR}_{si} = \frac{\sigma}{\sigma - 1} \left[\frac{R}{\alpha_s} \frac{1 + \tau_{Ksi}}{1 - \tau_{Ysi}} \right]^{\alpha_s} \left[\frac{1}{1 - \alpha_s} \frac{1}{1 - \tau_{Ysi}} \right]^{1 - \alpha_s}.$$

2. Collect terms

$$\text{TFPR}_{si} = \underbrace{\frac{\sigma}{\sigma - 1} \left(\frac{R}{\alpha_s} \right)^{\alpha_s} \left(\frac{1}{1 - \alpha_s} \right)^{1 - \alpha_s}}_{C_s} \frac{(1 + \tau_{Ksi})^{\alpha_s}}{(1 - \tau_{Ysi})^{\alpha_s + (1 - \alpha_s)}}.$$

$$\boxed{\text{TFPR}_{si} = C_s \frac{(1 + \tau_{Ksi})^{\alpha_s}}{1 - \tau_{Ysi}}}.$$

Capital wedge affects only the capital margin. Output wedge affects both capital and labor margins, so its exponent is one.

TFPQ versus TFPR: the key intuition

TFPQ: physical productivity

$$\text{TFPQ}_{si} = A_{si} = \frac{Y_{si}}{K_{si}^{\alpha_s} (wL_{si})^{1-\alpha_s}}.$$

- high TFPQ means good technology;
- efficient high-TFPQ plants should be large;
- TFPQ need not be equalized.

TFPR: revenue productivity

$$\text{TFPR}_{si} = P_{si} A_{si} = \frac{P_{si} Y_{si}}{K_{si}^{\alpha_s} (wL_{si})^{1-\alpha_s}}.$$

- high TFPR means high marginal products;
- high-TFPR plants are likely too small;
- TFPR should be equalized in an efficient allocation.

Core intuition: high TFPQ is technology; high TFPR is a constrained-scale signal.

From plant TFPR to industry TFP

1. TFPR links price and productivity

$$\text{TFPR}_{si} \equiv P_{si} A_{si} \Rightarrow P_{si} = \frac{\text{TFPR}_{si}}{A_{si}}.$$

2. CES price index

$$P_s = \left(\sum_i P_{si}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}.$$
$$P_s^{-1} = \left[\sum_i \left(\frac{A_{si}}{\text{TFPR}_{si}} \right)^{\sigma-1} \right]^{\frac{1}{\sigma-1}}.$$

A constrained high-productivity plant enters industry TFP as $A_{si} \times (\overline{\text{TFPR}}_s / \text{TFPR}_{si})$, so high TFPR reduces its effective weight.

3. Industry TFP

$$\text{TFP}_s = \left[\sum_{i=1}^{M_s} \left(A_{si} \frac{\overline{\text{TFPR}}_s}{\text{TFPR}_{si}} \right)^{\sigma-1} \right]^{\frac{1}{\sigma-1}}$$

Efficient benchmark

$$\text{TFPR}_{si} = \overline{\text{TFPR}}_s \Rightarrow \text{TFP}_s^{\text{eff}} = \left(\sum_i A_{si}^{\sigma-1} \right)^{\frac{1}{\sigma-1}}.$$

Counterfactual output and lognormal intuition

Sector efficiency ratio

$$\bar{A}_s \equiv \text{TFP}_s^{\text{eff}} = \left(\sum_i A_{si}^{\sigma-1} \right)^{\frac{1}{\sigma-1}}, \quad \Delta_s \equiv \frac{\text{TFP}_s}{\text{TFP}_s^{\text{eff}}}.$$

$$\Delta_s = \left[\sum_i \left(\frac{A_{si}}{\bar{A}_s} \frac{\overline{\text{TFPR}}_s}{\text{TFPR}_{si}} \right)^{\sigma-1} \right]^{\frac{1}{\sigma-1}}.$$

Aggregate output ratio

$$\frac{Y}{Y^{\text{eff}}} = \prod_s \Delta_s^{\theta_s}.$$

Closed-form intuition joint lognormal case

$$\log \text{TFP}_s = \log \text{TFP}_s^{\text{eff}} - \frac{\sigma}{2} \text{Var}(\log \text{TFPR}_{si})$$

More TFPR dispersion lowers Δ_s and therefore lowers Y/Y^{eff} .
The penalty is larger when σ is larger.

HK empirical payoff

TFPR dispersion and TFP gains

	TFPR 75/25	U.S.-efficiency TFP gain
China	2.3	30–50%
India	2.2	40–60%
United States	1.7	benchmark

Latest sample years; gains are relative to U.S. efficiency.

What the numbers mean

- China and India have wider marginal-product gaps.
- Reallocation toward high-TFPR plants raises TFP.
- Exact values depend on measurement and model assumptions.

Bottom line: aggregate productivity depends on where resources go, not only on plant technology.

Why the U.S. benchmark matters

Perfect equalization

$$\text{TFPR}_{si} = \overline{\text{TFPR}_s} \quad \forall i.$$

- clean theoretical benchmark;
- treats all TFPR dispersion as distortion;
- may overstate gains if data are noisy.

U.S.-efficiency benchmark

China/India dispersion $\rightarrow$ U.S. dispersion.

- allows normal dispersion;
- reduces measurement-error concern;
- gives a more conservative estimate.

HK's result is best read as *potential gains from reducing excess dispersion*, not the causal effect of one policy.

Part IV. Interpreting dispersion carefully

Dynamic inputs, capital adjustment, and static MRPK gaps

Asker et al.: the caveat to static misallocation

Question. When we observe MRPK dispersion, is it always evidence of policy misallocation?

HK static interpretation

$MRPK_i \neq MRPK_j \Rightarrow$ capital is not efficiently allocated.

- capital and labor can be reallocated within the period;
- marginal products should be equalized;
- persistent gaps look like wedges.

Asker dynamic interpretation

K_t chosen before today's shock

$\Rightarrow$ $MRPK_t$ can differ across firms.

- capital is a dynamic input;
- adjustment is costly and slow;
- static gaps can be dynamically optimal.

Main message: dispersion is a useful signal, but not automatic proof of inefficient policy wedges.

Asker et al.: the mechanism

1. Sales and revenue productivity

$$S_{it} = \Omega_{it} K_{it}^{\beta_K} L_{it}^{\beta_L} M_{it}^{\beta_M}, \quad q_{it} \equiv \log \Omega_{it}.$$

2. Measured static MRPK

$$\boxed{\text{MRPK}_{it} = \log(\beta_K) + s_{it} - k_{it}}, \quad s_{it} = \log S_{it}, \quad k_{it} = \log K_{it}.$$

3. TFPR shock and capital law

$$q_{it} = \mu + \rho q_{it-1} + \sigma_q \varepsilon_{it}, \quad K_{it+1} = (1 - \delta)K_{it} + I_{it}.$$

Mechanism

$$q_{it} \uparrow \Rightarrow S_{it} \uparrow$$

K_{it} predetermined $\Rightarrow k_{it}$ does not jump

$$\Rightarrow s_{it} - k_{it} \uparrow \Rightarrow \text{MRPK}_{it} \uparrow.$$

Key notation. i : producer; t : time; S : sales; Ω : revenue productivity; q : log TFPR; K, L, M : capital, labor, materials; I : investment; δ : depreciation; ρ : persistence; σ_q : volatility.

Intuition. A positive TFPR shock raises sales immediately, but capital is partly predetermined. Since measured MRPK is log sales minus log capital, MRPK can rise temporarily even without a policy wedge.

Asker et al.: adjustment costs and predictions

Dynamic investment problem

$$V(Q, K) = \max_I \{ \pi(Q, K) - C(I, K, Q) + \beta E[V(Q', (1 - \delta)K + I) \mid Q] \}.$$

Adjustment cost

$$C(I, K, Q) = I + C_K^F \mathbf{1}\{I \neq 0\} \pi(Q, K) + C_K^Q K \left(\frac{I}{K} \right)^2.$$

Fixed and convex costs make capital respond slowly to shocks.

How to read this. The model does not say misallocation is unimportant. It says a static reallocation exercise can overstate welfare gains if some dispersion is generated by dynamic adjustment.

Prediction

$$\rho = 0 \quad \Rightarrow \quad \boxed{\text{SD}(\text{MRPK}) \approx \frac{1}{\beta_K + \varepsilon^{-1}} \sigma_q}.$$

More volatile TFP shocks generate more static MRPK dispersion when capital cannot adjust instantly.

Empirical pattern

$$\text{TFPR volatility} \uparrow \quad \Rightarrow \quad \text{MRPK dispersion} \uparrow.$$

Connecting Asker back to HK: labor–capital wedges

HK static wedge interpretation

$$\text{MRPL}_{si} = \frac{w}{1 - \tau_{Ysi}}, \quad \text{MRPK}_{si} = R \frac{1 + \tau_{Ksi}}{1 - \tau_{Ysi}}.$$

- output wedge affects both capital and labor margins;
- capital wedge affects capital relative to labor;
- TFPR summarizes the combined wedge.

Asker dynamic caveat

TFPR shock today $\Rightarrow S_{it} \uparrow$, K_{it} slow.
 $\Rightarrow \text{MRPK}_{it} \uparrow$ even without a policy wedge.

- labor and materials are treated as more flexible;
- capital is the main dynamic input;
- measured capital wedge may include adjustment dynamics.

Clean takeaway. Asker et al. motivate joint interpretation of labor and capital margins: observed wedges are reduced-form equilibrium objects, not pure policy taxes. A high MRPK may reflect credit frictions, but it may also reflect slow capital adjustment after a TFPR shock.

Part IV. Synthesis

What the literature teaches us around HK

Putting the papers together

Paper	Key object	Research role	Main caution
OP (1996)	$P_t = \bar{p}_t + \sum_i \Delta s_i \Delta p_i$	Reallocation component of productivity growth	Need consistent productivity estimates.
RR (2008)	$k, n \propto [(1 - \tau)s]^{1/(1-\alpha-\gamma)}$	Policies distort establishment size and TFP	Model experiment is abstract.
HK (2009)	$TFPR_i \propto (1 + \tau_{Ki})^\alpha / (1 - \tau_{Yi})$	Measure wedges and counterfactual TFP gains	TFPR includes noise, markups, and dynamics.
Asker et al. (2014)	SD(MRPK) $\uparrow$ with TFPR volatility	Dynamic capital adjustment component	Static gaps are not always inefficient.

Technology, allocation, policy, and dynamics jointly determine aggregate productivity.

How to interpret observed dispersion

Observed dispersion is not the conclusion; it is the starting point.

What dispersion can mean

- policy distortions,
- credit frictions,
- markups and demand differences,
- adjustment costs,
- measurement error,
- entry and exit selection.

How the papers discipline it

- OP: estimate productivity and decompose reallocation.
- RR: simulate distortions in a GE model.
- HK: infer wedges from TFPR and compute gains.
- Asker: separate dynamic adjustment from static inefficiency.

TFPR dispersion is a useful diagnostic, but welfare interpretation depends on the mechanism.

Final takeaway

Aggregate productivity is not only about *which technologies exist*; it is also about *who gets the resources*.

Measurement

$$P_t = \bar{p}_t + \text{covariance}$$

OP: reallocation is visible in plant data.

Misallocation

$$TFP_s = \left[\sum_i (A_i \cdot \text{weight}_i)^{\sigma-1} \right]^{1/(\sigma-1)}$$

HK/RR: wedges reduce aggregate TFP.

Dynamics

$$K_{t+1} = (1 - \delta)K_t + I_t$$

Asker: some gaps reflect slow adjustment.

Central message: productive firms must be allowed to become large.

Backup: notation dictionary

HK and RR notation

Y, Y_s, Y_{si}	final, industry, and plant output
P_s, P_{si}	industry price index and plant price
A_{si}, s	plant or establishment productivity
K, L, k, n	capital and labor inputs
w, R, r	wage and capital rental rate
α_s, α, γ	production elasticities
σ	elasticity of substitution
τ_Y, τ_K, τ	output, capital, or producer wedge
MRPK, MRPL	marginal revenue products
TFPQ, TFPR	physical and revenue productivity

OP and Asker notation

y, s, k, l, m	log output or sales, capital, labor, materials
ω, q	unobserved productivity or TFPR shock
η	measurement or unexpected shock
i, I	investment
a	plant age
χ	survival indicator
$V(\cdot)$	dynamic value function
$C(I, K, Q)$	capital adjustment cost
ρ, σ_q	persistence and volatility of TFPR
P_t, p_{it}, s_{it}	industry productivity, plant productivity, output share